

Unit 4.4 Indeterminate

Note Title

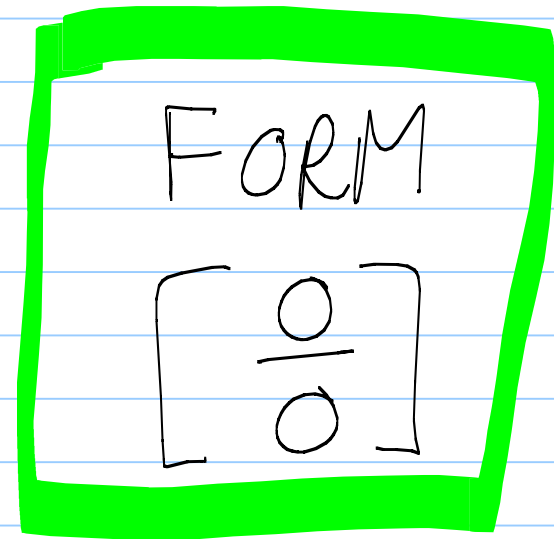
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forms and L'Hospital Rule.

Stewart, Section 4.4 p301-309

Revision

$$\bullet \lim_{x \rightarrow 2} \frac{x-2}{x^2-4}$$



$$= \lim_{x \rightarrow 2} \frac{x-2}{(x-2)(x+2)} = \lim_{x \rightarrow 2} \frac{1}{x+2} = \frac{1}{4}.$$

$$\bullet \lim_{x \rightarrow \infty} \frac{x^5 + 5x^3}{x^4 + x^2}$$

FORM
 $\left[\frac{\infty}{\infty} \right]$

$$= \lim_{x \rightarrow \infty} \frac{x^4 (x + 5/x)}{x^4 (1 + \frac{1}{x^2})}$$

$$= \lim_{x \rightarrow \infty} \frac{x + 5/x}{1 + 1/x^2} = \infty$$

Consider

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

if $f(x) \rightarrow 0$ & $g(x) \rightarrow 0$ as $x \rightarrow a$

we have the indeterminate

form of type $\frac{0}{0}$

if $f(x) \rightarrow \infty$ & $g(x) \Rightarrow \infty$ as $x \rightarrow \infty$

we have the indeterminate
form of type $\frac{\infty}{\infty}$.

There are 7 indeterminate forms :

1. $\frac{0}{0}$, 2. $\frac{+\infty}{+\infty}$

3. $\infty - \infty$ Indeterminate difference

4. $0 \times \infty$ Indeterminate products

5.

$$0^0$$

6.

$$\infty^0$$

7.

$$1^\infty$$

Indeterminate
powers.

L'Hospital Rule p302.

Suppose f & g are differentiable and $g'(x) \neq 0$ on an open interval I that contains a (except possibly at a). Suppose that

$$\lim_{x \rightarrow a} f(x) = 0 \text{ and } \lim_{x \rightarrow a} g(x) = 0$$

or that

$$\lim_{x \rightarrow a} f(x) = \pm \infty \text{ and } \lim_{x \rightarrow a} g(x) = \pm \infty$$

i.e. we have the indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$, Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

if the limit on the right exists

Example 1

$$\lim_{x \rightarrow 2} \frac{x-2}{x^2-4}$$

FORM

$$\left[\frac{0}{0} \right]$$

$$\stackrel{\text{L'Hop}}{=} \lim_{x \rightarrow 2} \frac{1}{2x} = \frac{1}{4}$$

Exemple 2

Solve:

$$\textcircled{1} \lim_{x \rightarrow \infty} \frac{2^x}{x^2}$$

FORM
 $\left[\frac{\infty}{\infty} \right]$

$$\stackrel{\text{L'Hop}}{=} \lim_{x \rightarrow \infty} \frac{\ln 2 \cdot 2^x}{2x}$$

FORM $\left[\frac{\infty}{\infty} \right]$

$$\stackrel{\text{L'Hop}}{=} \lim_{x \rightarrow \infty} \frac{\ln 2 \cdot \ln 2 \cdot 2^x}{2} = \infty$$

②

$$\lim_{x \rightarrow 1} \frac{\ln x}{x-1}$$

FORM
[0/0]

L'Hop
=

$$\lim_{x \rightarrow 1} \frac{\frac{1}{x}}{1} = 1$$

$$\textcircled{3} \lim_{x \rightarrow \infty} \frac{\ln(\ln x)}{\ln x}$$

FoRM
[$\frac{\infty}{\infty}$]

$$\stackrel{\text{L'Hop}}{=} \lim_{x \rightarrow \infty} \frac{\frac{1}{\ln x} \cdot \frac{1}{x}}{\frac{1}{x}}$$

$$= 1$$

$$\textcircled{4} \quad \lim_{x \rightarrow \infty} \frac{e^x}{x^2}$$

FORM
 $\left[\frac{\infty}{\infty} \right]$

$$\stackrel{\text{L'Hop}}{=} \lim_{x \rightarrow \infty} \frac{e^x}{2x}$$

FORM
 $\left[\frac{\infty}{\infty} \right]$

$$\stackrel{\text{L'Hop}}{=} \lim_{x \rightarrow \infty} \frac{e^x}{2} = \infty$$

FORM $\infty - \infty$ (try convert
the difference
into a quotient)

Examples

$$1. \lim_{x \rightarrow \infty} x^3 - x^4$$

Form
($\infty - \infty$)

$$= \lim_{x \rightarrow \infty} x^3(1-x) = -\infty$$

$$2. \lim_{x \rightarrow -\infty} x^4 + x^3$$

Form
($\infty - \infty$)

$$= \lim_{x \rightarrow -\infty} x^3 (x + 1)$$

$$= \infty$$

$$3. \lim_{cc \rightarrow \infty} cc^3 + cc^4 = \infty$$

$$4) \lim_{x \rightarrow 1^+} \left(\frac{1}{\ln x} - \frac{1}{x-1} \right) \quad \text{Form } (\infty - \infty)$$

$$= \lim_{x \rightarrow 1^+} \left(\frac{(x-1) - \ln x}{\ln x (x-1)} \right) \quad \text{Form } \left(\frac{0}{0} \right)$$

$$\stackrel{\text{L'Hop}}{=} \lim_{x \rightarrow 1^+} \frac{1 - \frac{1}{x}}{\frac{1}{x}(x-1) + \ln x (1)}$$

$$= \lim_{x \rightarrow 1^+} \frac{\frac{x-1}{x}}{\left(1 - \frac{1}{x} + \ln x \right)}$$

$$= \lim_{x \rightarrow 1^+} \frac{x-1}{x-1+x \ln x}$$

Form
 $\left(\frac{0}{0}\right)$

L'Hop

$$= \lim_{x \rightarrow 1^+} \frac{1}{1 + \ln x + x \cdot \frac{1}{x}}$$

$$= \lim_{x \rightarrow 1^+} \frac{1}{2 + \ln x} = \frac{1}{2}$$

FORM $\infty \times 0$

If $\lim_{x \rightarrow a} f(x) = 0$ and

$$\lim_{x \rightarrow a} g(x) = \infty$$

$$\lim_{x \rightarrow a} f(x) \cdot g(x)$$

FORM
 $0 \times \infty$

then write the product as a quotient

$$fg = \frac{f}{1/g} \quad \text{or} \quad \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} \infty \\ \infty \end{bmatrix}$$

$$fg = g / 1/f$$

and use L'Hopital's Rule

Examples

$$\lim_{x \rightarrow -\infty} x e^x \quad \text{Form } (-\infty \cdot 0)$$

$$= \lim_{x \rightarrow -\infty} \frac{x}{e^{-x}} \quad \text{Form } \left(\frac{-\infty}{\infty} \right)$$

$$= \lim_{x \rightarrow -\infty} \frac{1}{-e^{-x}}$$

$$= 0$$

$$2) \lim_{x \rightarrow 0^+} x \ln x$$

Form
(0 · -∞)

$$= \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}}$$

$$= \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}}$$

$$= \lim_{x \rightarrow 0^+} \frac{1}{x} \cdot -\frac{1}{x^2}$$

$$= \lim_{x \rightarrow 0^+} -\frac{1}{x} \times \frac{x^2}{1}$$

$$= \lim_{x \rightarrow 0^+} -x$$

$$= 0$$

Homework.

① $\lim_{x \rightarrow \infty} x \sin \frac{1}{x}$

② $\lim_{x \rightarrow \infty} x e^{3/x}$

(Not an indeterminate form)

FORMS 0^0 , ∞^0 , 1^∞

$$\lim_{x \rightarrow a} f(x)^{g(x)}.$$

Use $\ln$ function

$$\text{Let } y = f(x)^{g(x)}$$

$$\ln y = \ln f(x)^{g(x)}$$

• Simplify:

$$\ln y = g(x) \ln f(x)$$

• $\lim_{x \rightarrow a} g(x) \ln f(x)$ FORM $[0 \cdot \infty]$

$= \lim_{x \rightarrow a} \frac{\ln f(x)}{1/g(x)}$ FORM $\left[\frac{\infty}{\infty} \right]$

Then apply L'Hop.

Remember we want the limit of y not $\ln y$ so

$$\lim_{x \rightarrow a} y = \lim_{x \rightarrow a} e^{\ln y} = e^{\lim_{x \rightarrow a} \ln y}$$

Example Find

$$\lim_{x \rightarrow 0^+} x^x \quad \text{Form } [0^0]$$

$$\text{Let } y = x^x$$

$$\ln y = x \ln x$$

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} x \ln x$$

Form
[0 · -∞]

$$= \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}}$$

$$= 0$$

$$\lim_{x \rightarrow 0^+} y = \lim_{x \rightarrow 0^+} e^{\ln y}$$

$$= e^{\lim_{x \rightarrow 0^+} \ln y}$$

$$= e^0 = 1$$

Examples

$$\textcircled{1} \quad \lim_{x \rightarrow \infty} \left(1 + \frac{4}{x}\right)^{3x} \quad \text{Form } [1, \infty]$$

$$\text{Let } y = \left(1 + \frac{4}{x}\right)^{3x}$$

$$\begin{aligned} \ln y &= \ln \left(1 + \frac{4}{x}\right)^{3x} \\ &= 3x \ln \left(1 + \frac{4}{x}\right) \end{aligned}$$

$$\lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} 3x \ln\left(1 + \frac{4}{x}\right) \quad \text{Form } (\infty \cdot 0)$$

$$= \lim_{x \rightarrow \infty} \frac{3 \ln\left(1 + \frac{4}{x}\right)}{\frac{1}{x}} \quad \text{Form } \left(\frac{0}{0}\right)$$

$$= \lim_{x \rightarrow \infty} 3 \cdot \frac{1}{1 + \frac{4}{x}} \left(-\frac{4}{x^2}\right)$$

$$-\frac{1}{x^2}$$

$$= \lim_{x \rightarrow \infty} \frac{3 \cdot 4}{1 + \frac{4}{x}}$$

$$= 12$$

$$\lim_{x \rightarrow \infty} y = \lim_{x \rightarrow \infty} e^{\ln y}$$

$$= e^{\lim_{x \rightarrow \infty} \ln y} = e^{12}$$

$$\textcircled{2} \quad \lim_{x \rightarrow \infty} (e^x + 6x)^{\frac{2}{x}} \quad \text{Form } (\infty^0)$$

$$\text{Let } y = (e^x + 6x)^{2/x}$$

$$\begin{aligned} \ln y &= \ln (e^x + 6x)^{\frac{2}{x}} \\ &= \frac{2 \ln (e^x + 6x)}{x} \end{aligned}$$

$$\lim_{x \rightarrow \infty} \frac{2 \ln(e^x + 6x)}{x} \quad \left(\frac{\infty}{\infty} \right)$$

$$\stackrel{\text{L'Hop}}{=} \lim_{x \rightarrow \infty} 2 \cdot \frac{1}{e^x + 6x} (e^x + 6) \quad \left(\frac{\infty}{\infty} \right)$$

$$= \lim_{x \rightarrow \infty} 2 \cdot \frac{e^x}{e^x + 6} \quad \left(\frac{\infty}{\infty} \right)$$

$$= \lim_{x \rightarrow \infty} 2 \cdot \frac{e^x}{e^x} = \lim_{x \rightarrow \infty} 2 = 2$$

$$\therefore \lim_{x \rightarrow \infty} y = \lim_{x \rightarrow \infty} e^{hy} = e^2$$

$$\textcircled{3} \quad \lim_{x \rightarrow 0^+} x^{\frac{1}{\ln x}} \quad \begin{array}{l} \text{FORM} \\ (0^0) \end{array}$$

$$\text{Let } y = x^{\frac{1}{\ln x}}$$

$$\ln y = \frac{1}{\ln x} \cdot \ln x$$

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{\ln x} \quad \left(\frac{-\infty}{-\infty} \right)$$

$$= \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{\frac{1}{x}}$$

$$= 1$$

$$\therefore \lim_{x \rightarrow 0^+} y = e^1$$

Homework

$$1) \lim_{x \rightarrow 0^+} (1 + \sin x)^{\cot x} \quad [1^\infty]$$

$$2) \lim_{x \rightarrow 0^+} (e^{-\frac{1}{x}})^x \quad [0^0]$$

$$3) \lim_{x \rightarrow 0} (e^x + x)^{\frac{1}{x}}$$

$$4) \lim_{x \rightarrow 0} (\cos x)^{\frac{2}{x}}$$